A CENTRAL SIMPLE SUBALGEBRA MAKES THE ALGEBRA A TENSOR PRODUCT WITH ITS CENTRALIZER

THE STACKS PROJECT AUTHORS

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Editorial note. The proof is the author's complete argument; each of its two steps names the result it uses and the conclusion is drawn inside the span, so no step is left to the reader.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The degree formula for a centralizer. Same chapter, Theorem theorem-centralizer. Supplies that the degree of the algebra equals the product of the degrees of the subalgebra and of its centralizer, which gives the dimension equality used in the proof.
- Simplicity of the tensor product. Same chapter, Lemma lemma-tensor-simple. Supplies that the tensor product of the subalgebra with its centralizer is simple, so that the natural map into the algebra out of a simple algebra is injective.

Source reference. Original theorem 074T: sources/stacks-brauer/pinned-original-source.tex, source lines 746–746. Navigation pointer invoked at proof line 793; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 074F: sources/stacks-brauer/pinned-original-source.tex, source lines 538–538. Navigation pointer invoked at proof line 765; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Lemma 0.1. *Let A be a finite central simple algebra over k, and let B be a simple subalgebra of A. If B is a central k-algebra, then $A = B \otimes_k C$ where C is the (central simple) centralizer of B in A.*

Proof. We have $\dim_k(A) = \dim_k(B \otimes_k C)$ by Theorem 074T. By Lemma 074F the tensor product is simple. Hence the natural map $B \otimes_k C \to A$ is injective hence an isomorphism. □